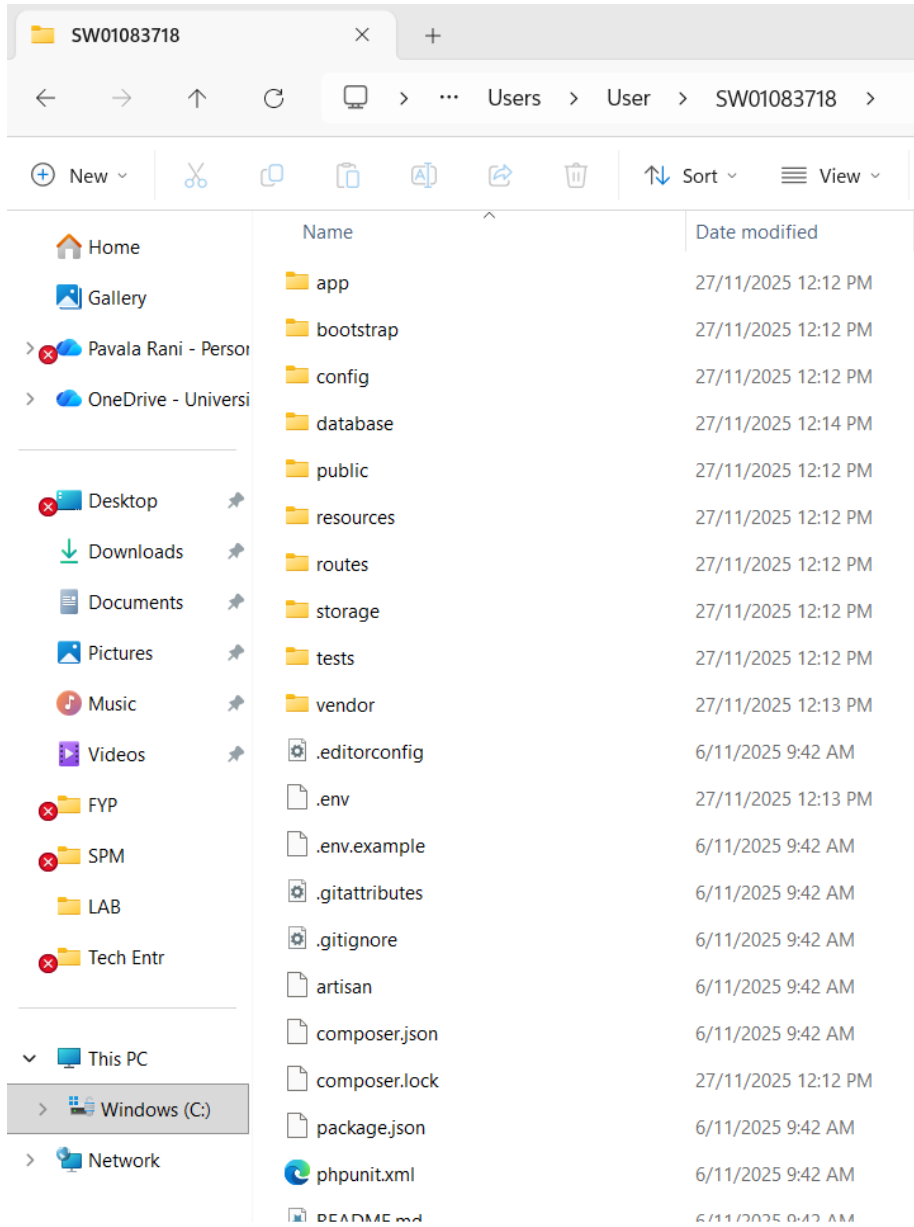


CSEB5123 Advanced Web Application Development

LAB TEST

Student ID: SW01083718 **Name:** Pavala Rani A/P Sunderan **Section:** 01TB

Q1 : Project Directory Structure



SW01083718		
Users > User > SW01083718		
New	Name	Date modified
	app	27/11/2025 12:12 PM
	bootstrap	27/11/2025 12:12 PM
	config	27/11/2025 12:12 PM
	database	27/11/2025 12:14 PM
	public	27/11/2025 12:12 PM
	resources	27/11/2025 12:12 PM
	routes	27/11/2025 12:12 PM
	storage	27/11/2025 12:12 PM
	tests	27/11/2025 12:12 PM
	vendor	27/11/2025 12:13 PM
	.editorconfig	6/11/2025 9:42 AM
	.env	27/11/2025 12:13 PM
	.env.example	6/11/2025 9:42 AM
	.gitattributes	6/11/2025 9:42 AM
	.gitignore	6/11/2025 9:42 AM
	artisan	6/11/2025 9:42 AM
	composer.json	6/11/2025 9:42 AM
	composer.lock	27/11/2025 12:12 PM
	package.json	6/11/2025 9:42 AM
	phpunit.xml	6/11/2025 9:42 AM
	README.md	6/11/2025 9:42 AM

Q2 : Artisan Command Output

```
C:\Users\User\SW01083718>php artisan migrate

  INFO  Preparing database.

Creating migration table ..... 409.82ms DONE

  INFO  Running migrations.

0001_01_01_000000_create_users_table ..... 365.24ms DONE
0001_01_01_000001_create_cache_table ..... 37.60ms DONE
0001_01_01_000002_create_jobs_table ..... 134.44ms DONE
```

Q3.1 : Artisan command output (create model, controller and migration file)

```
C:\Users\User\SW01083718>php artisan make:model Book -m

  INFO  Model [C:\Users\User\SW01083718\app\Models\Book.php] created successfully.

  INFO  Migration [C:\Users\User\SW01083718\database\Migrations\2025_11_27_043102_create_books_table.php] created successfully.

C:\Users\User\SW01083718>php artisan make:model Product -m

  INFO  Model [C:\Users\User\SW01083718\app\Models\Product.php] created successfully.

  INFO  Migration [C:\Users\User\SW01083718\database\Migrations\2025_11_27_043111_create_products_table.php] created successfully.

C:\Users\User\SW01083718>php artisan make:controller BookController --resource

  INFO  Controller [C:\Users\User\SW01083718\app\Http\Controllers\BookController.php] created successfully.

C:\Users\User\SW01083718>php artisan make:controller ProductController --resource

  INFO  Controller [C:\Users\User\SW01083718\app\Http\Controllers\ProductController.php] created successfully.
```

Q3.2 : Output of artisan command route:list

```
C:\Users\User\SW01083718>php artisan route:list

GET|HEAD      / .....
GET|HEAD      products ..... products.index > ProductController@index
POST          products ..... products.store > ProductController@store
GET|HEAD      products/create products.create > ProductController@cre...
GET|HEAD      products/{product} products.show > ProductController@sh...
PUT|PATCH    products/{product} products.update > ProductController@...
DELETE        products/{product} products.destroy > ProductController...
GET|HEAD      products/{product}/edit products.edit > ProductControll...
GET|HEAD      storage/{path} ..... storage.local
GET|HEAD      up .....

Showing [10] routes
```

Q4.1 : Content of the respective migration file

```
2025_11_27_043111_create_products_table.php X
database > migrations > 2025_11_27_043111_create_products_table.php
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      /**
10       * Run the migrations.
11       */
12     public function up(): void
13     {
14         Schema::create('products', function (Blueprint $table) {
15             $table->id();
16             $table->string('name');
17             $table->string('barcode');
18             $table->foreignId('supplier_id')
19                 ->constrained()
20                 ->onDelete('cascade');
21             $table->timestamps();
22         });
23     }
24 }
```

Q4.2 : Output of artisan migration command

```
C:\Users\User\SW01083718>php artisan migrate:fresh

Dropping all tables ..... 123.66ms DONE
INFO Preparing database.
Creating migration table ..... 17.79ms DONE
INFO Running migrations.

0001_01_01_000000_create_users_table ..... 100.41ms DONE
0001_01_01_000001_create_cache_table ..... 21.04ms DONE
0001_01_01_000002_create_jobs_table ..... 99.15ms DONE
2025_11_27_043000_create_suppliers_table ..... 12.89ms DONE
2025_11_27_043102_create_books_table ..... 13.21ms DONE
2025_11_27_043111_create_products_table ..... 78.39ms DONE
```

Q5.1 : Artisan command output (create model)

```
C:\Users\User\SW01083718>php artisan make:model Supplier -m

  INFO  Model [C:\Users\User\SW01083718\app\Models\Supplier.php] created successfully.

  INFO  Migration [C:\Users\User\SW01083718\database\Migrations\2025_11_27_044004_create_suppliers_table.php] created successfully.
```

```
C:\Users\User\SW01083718>php artisan make:controller SupplierController --resource

  INFO  Controller [C:\Users\User\SW01083718\app\Http\Controllers\SupplierController.php] created successfully.
```

Q5.2 : Output of artisan command route:list

```
C:\Users\User\SW01083718>php artisan route:list

GET|HEAD      / .....
GET|HEAD      products ..... products.index > ProductController@index
POST          products ..... products.store > ProductController@store
GET|HEAD      products/create products.create > ProductController@create
GET|HEAD      products/{product} products.show > ProductController@show
PUT|PATCH    products/{product} products.update > ProductController@update
DELETE        products/{product} products.destroy > ProductController@destroy
GET|HEAD      products/{product}/edit products.edit > ProductController@edit
GET|HEAD      storage/{path} ..... storage.local
GET|HEAD      suppliers ... suppliers.index > SupplierController@index
POST          suppliers ... suppliers.store > SupplierController@store
GET|HEAD      suppliers/create suppliers.create > SupplierController@create
GET|HEAD      suppliers/{supplier} suppliers.show > SupplierController@show
PUT|PATCH    suppliers/{supplier} suppliers.update > SupplierController@update
DELETE        suppliers/{supplier} suppliers.destroy > SupplierController@destroy
GET|HEAD      suppliers/{supplier}/edit suppliers.edit > SupplierController@edit
GET|HEAD      up .....

Showing [17] routes
```

Q6.1 : Content of the respective migration file

```
2025_11_27_044004_create_suppliers_table.php X
database > migrations > 2025_11_27_044004_create_suppliers_table.php
1  <?php
2
3  use Illuminate\Database\Migrations\Migration;
4  use Illuminate\Database\Schema\Blueprint;
5  use Illuminate\Support\Facades\Schema;
6
7  return new class extends Migration
8  {
9      /**
10       * Run the migrations.
11       */
12     public function up(): void
13     {
14         Schema::create('suppliers', function (Blueprint $table) {
15             $table->id();
16             $table->string('name');
17             $table->string('address');
18             $table->string('contactNo');
19             $table->timestamps();
20         });
21     }
22
23     /**
24      * Reverse the migrations.
25      */
26     public function down(): void
27     {
28         Schema::dropIfExists('suppliers');
29     }
30 };
```

Q6.2 : Output of artisan migration command

```
C:\Users\User\SW01083718>php artisan migrate

INFO Running migrations.

2025_11_27_043102_create_books_table ..... 21.70ms DONE
2025_11_27_043111_create_products_table ..... 11.89ms DONE
2025_11_27_044004_create_suppliers_table ..... 11.80ms DONE
```

Q7.1 : Relationship function [supplier.php]

```
Supplier.php ●
app > Models > Supplier.php
1  <?php
2
3  namespace App\Models;
4  use Illuminate\Database\Eloquent\Model;
5  use App\Models\Product;
6
7  class Supplier extends Model
8  {
9      use HasFactory;
10     protected $fillable = [
11         'name',
12         'address',
13         'contactNo',
14     ];
15     public function products()
16     {
17         return $this->hasMany(Product::class);
18     }
19 }
```

Q7.2 : Relationship function [product.php]

```
Product.php ✕
app > Models > Product.php
1  <?php
2
3  namespace App\Models;
4  use Illuminate\Database\Eloquent\Factories\HasFactory;
5  use Illuminate\Database\Eloquent\Model;
6  use App\Models\Supplier;
7
8  class Product extends Model
9  {
10     use HasFactory;
11     protected $fillable = [
12         'name',
13         'barcode',
14         'supplier_id',
15     ];
16     public function supplier()
17     {
18         return $this->belongsTo(Supplier::class);
19     }
20 }
```

Q8.1 : CRUD methods in ProductController

```
<?php

namespace App\Http\Controllers;

use App\Models\Product;
use App\Models\Supplier;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();
        return view('products.index', compact('products'));
    }

    public function create()
    {
        $suppliers = Supplier::all();
        return view('products.create', compact('suppliers'));
    }

    public function store(Request $request)
    {
        Product::create([
            'name' => $request->name,
            'barcode' => $request->barcode,
            'supplier_id' => $request->supplier_id,
        ]);

        return redirect()->route('products.index');
    }

    public function show(Product $product)
    {
        return view('products.show', compact('product'));
    }
}
```

```
public function edit(Product $product)
{
    $suppliers = Supplier::all();
    return view('products.edit', compact('product', 'suppliers'));
}

public function update(Request $request, Product $product)
{
    $product->update([
        'name'      => $request->name,
        'barcode'   => $request->barcode,
        'supplier_id' => $request->supplier_id,
    ]);
    return redirect()->route('products.index');
}

public function destroy(Product $product)
{
    $product->delete();
    return redirect()->route('products.index');
}
}
```


Q8.2 : CRUD methods in SupplierController

```
<?php
namespace App\Http\Controllers;
use App\Models\Supplier;
use Illuminate\Http\Request;

class SupplierController extends Controller
{
    public function index()
    {
        $suppliers = Supplier::all();
        return view('suppliers.index', compact('suppliers'));
    }

    public function create()
    {
        return view('suppliers.create');
    }

    public function store(Request $request)
    {
        Supplier::create([
            'name'      => $request->name,
            'address'   => $request->address,
            'contactNo' => $request->contactNo,
        ]);
        return redirect()->route('suppliers.index');
    }

    public function show(Supplier $supplier)
    {
        return view('suppliers.show', compact('supplier'));
    }

    public function edit(Supplier $supplier)
    {
        return view('suppliers.edit', compact('supplier'));
    }
}
```

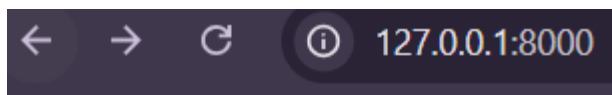
```

public function update(Request $request, Supplier $supplier)
{
    $supplier->update([
        'name'      => $request->name,
        'address'   => $request->address,
        'contactNo' => $request->contactNo,
    ]);
    return redirect()->route('suppliers.index');
}

public function destroy(Supplier $supplier)
{
    $supplier->delete();
    return redirect()->route('suppliers.index');
}
}

```

Q9.1 : Application main page with links to Product & Supplier



Welcome

- [Product List](#)
- [Supplier List](#)

Q9.2 : Product Views

9.2.1 : list all products

All Products

[Create New Product](#) | [View Suppliers](#)

ID	Name	Barcode	Supplier	Actions
1	orange	11111111111111111111	chua	Show Edit <button>Delete</button>
3	apple	33333333333333333333	rosh	Show Edit <button>Delete</button>

9.2.2 : create new product

Create New Product

Product Name:

Barcode:

Supplier:

[Back to Product List](#)

9.2.3 : edit product

Edit Product

Product Name:

Barcode:

Supplier:

[Back to Product List](#)

9.2.4 : show product with its supplier

Product Details

ID: 1

Name: orange

Barcode: 11111111111111111111

Supplier

Name: chua

Address: kedah

Contact No: 01243564844

[Back to Product List](#)

Q9.3 : Supplier Views

9.3.1 : list all suppliers

All Suppliers				
Create New Supplier View Products				
ID	Name	Address	Contact No	Actions
1	rosh	jalan watan 2	01123456789	Show Edit <button>Delete</button>
2	chua	kedah	01243564844	Show Edit <button>Delete</button>
3	pav	jalan watan 2	123456789	Show Edit <button>Delete</button>

9.3.2 : create new supplier

Create New Supplier	
Name:	
Address:	
Contact No:	
	<button>Save</button>
Back to Supplier List	

9.3.3 : edit supplier

Edit Supplier	
Name:	rosh
Address:	jalan watan 2
Contact No:	01123456789
	<button>Update</button>
Back to Supplier List	

9.3.4 : show supplier's products

Supplier Details

ID: 2

Name: chua

Address: kedah

Contact No: 01243564844

Products supplied by this supplier

- orange (Barcode: 11111111111111111111)

[Back to Supplier List](#)